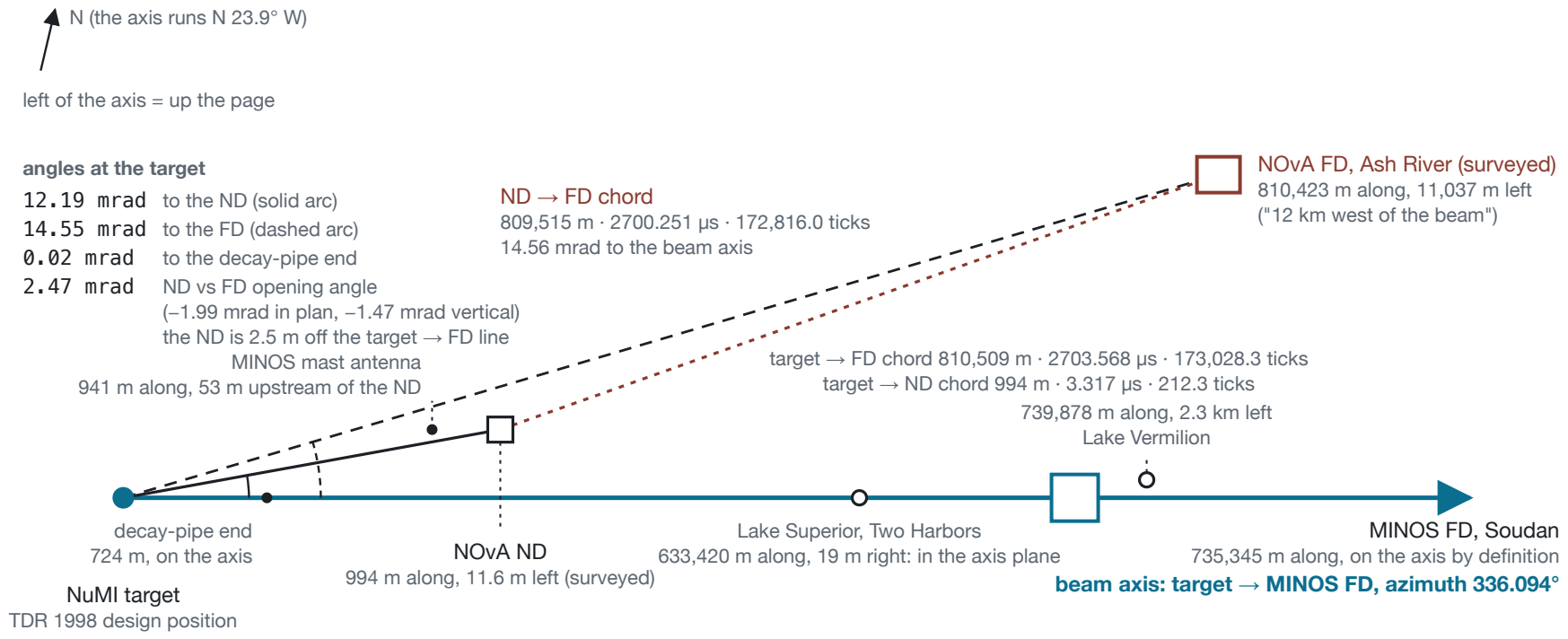

N (the axis runs N 23.9° W)
left of the axis = up the page
angles at the target
12.19 mrad to the ND (solid arc)
14.55 mrad to the FD (dashed arc)
0.02 mrad to the decay-pipe end
2.47 mrad ND vs FD opening angle
(–1.99 mrad in plan, –1.47 mrad vertical)
the ND is 2.5 m off the target → FD line
MINOS mast antenna
941 m along, 53 m upstream of the ND
ND → FD chord
809,515 m · 2700.251 μs · 172,816.0 ticks
14.56 mrad to the beam axis
NOvA FD, Ash River (surveyed)
810,423 m along, 11,037 m left
("12 km west of the beam")
target → FD chord 810,509 m · 2703.568 μs · 173,028.3 ticks
target → ND chord 994 m · 3.317 μs · 212.3 ticks
739,878 m along, 2.3 km left
Lake Vermilion
decay-pipe end
724 m, on the axis
NuMI target
TDR 1998 design position
NOvA ND
994 m along, 11.6 m left (surveyed)
Lake Superior, Two Harbors
633,420 m along, 19 m right: in the axis plane
MINOS FD, Soudan
735,345 m along, on the axis by definition
beam axis: target → MINOS FD, azimuth 336.094°